INCH-POUND
MIL-DTL-32572A(NAVY)
30 November 2022
SUPERSEDING
MIL-DTL-32572(NAVY)
21 February 2017

DETAIL SPECIFICATION

ANCHOR WINDLASS, ELECTRO-MECHANICAL, VERTICAL, TWO-SPEED AND VARIABLE-SPEED

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two-speed and variable-speed, electro-mechanical, vertical shaft anchor windlasses on naval ships.

1.2 Classification. Vertical axis, electro-mechanical anchor windlasses with one wildcat, one capstan head, and a below deck drive are available in the following types, as specified (see 6.2):

a. Type I – Two-speed system with drive components matched to the motor capabilities and system requirements.

b. Type II – Variable-speed with drive components matched to the motor-controlled torque and system requirements.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

FF-S-92 - Screw, Machine, Slotted, Cross-Recessed or Hexagon Head

RR-C-271 - Chains and Attachments, Carbon and Alloy Steel

COMMERCIAL ITEM DESCRIPTIONS

A-A-52522 - Lining Brake, and Lining Material Brake: Automotive (Metric)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 3950

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-917	-	Electric Power Equipment, Basic Requirements for
MIL-DTL-2036	-	Enclosures for Electric and Electronic Equipment, Naval Shipboard
MIL-P-15024/5	-	Plate, Identification
MIL-DTL-16392	-	Brake, Electro-Mechanical, Naval Shipboard
MIL-DTL-17060	-	Motors, Alternating Current, Integral-Horsepower, Shipboard Use
MIL-DTL-17361	-	Circuit Breaker Types AQB/NQB, Air, Electric, Low Voltage, Insulated Housing (Shipboard Use), General Specification for
MIL-DTL-17944	-	Capstans, Electric - for Shipboard Use
MIL-PRF-23236	-	Coating Systems for Ship Structures
MIL-DTL-24441	-	Paint, Epoxy-Polyamide, General Specification for
MIL-PRF-24635	-	Coating Systems, Weather-Resistant, Exterior Use
MIL-C-24707/5	-	Castings, Ductile Iron and Austenitic Ductile Iron
MIL-PRF-32006	-	Controller, Programmable, Naval Shipboard
MIL-PRF-32168	-	Variable Speed Drive System for Induction and Synchronous Machines
MIL-DTL-32613	-	Controller, Auxiliary-System, Naval Shipboard

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108	-	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-464	-	Electromagnetic Environmental Effects Requirements for Systems
MIL-STD-740-2	-	Structureborne Vibratory Acceleration Measurements Acceptance Criteria of Shipboard Equipment
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-882	-	System Safety
MIL-STD-1310	-	Shipboard Bonding, Grounding, and other Techniques for Electromagnetic Compatibility, Electromagnetic Pulse (EMP) Mitigation, and Safety
MIL-STD-1399-300-1	-	Low Voltage Electric Power, Alternating Current
MIL-STD-1472	-	Human Engineering
MIL-STD-1474	-	Noise Limits

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DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-299 - Cable Comparison Handbook Data Pertaining to Electric Shipboard Cable

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume 1

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-860303 - Hull Type Plan, Capstan and Gypsy Heads, Tabulation of Sizes

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

S9086-TW-STM-010/582 - NSTM Chapter 582, Mooring and Towing

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the TMIN without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/AGMA 945-A18 - Splines - Design and Application

ANSI/AGMA 2101-D04 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth

ANSI/AGMA 6001-E08 - Design and Selection of Components for Enclosed Gear Drives

ANSI/AGMA 6123-B06 - Design Manual for Enclosed Epicyclic Gear Drives

(Copies of these documents are available online at https://webstore.ansi.org/.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B17.1 - Keys and Keyseats

(Copies of this document are available online at www.asme.org.)

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AMERICAN WELDING SOCIETY (AWS)

AWS A2.4 - Standard Symbols for Welding, Brazing, and Nondestructive Examination

(Copies of this document are available online at www.aws.org.)

ASSOCIATION FOR MATERIALS PROTECTION AND PERFORMANCE

SSPC-SP 10 (WAB)/NACE No. 2 - Near-White Metal Wet Abrasive Blast Cleaning

(Copies of this document are available online at www.ampp.org.)

ASTM INTERNATIONAL

ASTM F765 - Standard Specification for Wildcats, Ship Anchor Chain

(Copies of this document are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 4568 - Ships and marine technology — Sea-going vessels — Windlasses and anchor capstans

(Copies of this document are available online at www.iso.org.)

SAE INTERNATIONAL

SAE AMS-STD-595	-	Colors Used in Government Procurement
SAE AS35411	-	Fittings, Lubrication
SAE AS50151	-	Connectors, Electrical, Circular Threaded, AN Type, General Specification for
SAE J661	-	Brake Lining Quality Test Procedure
SAE J866	-	Friction Coefficient Identification and Environmental Marking System for Brake Linings

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials. The materials shall be capable of meeting all of the requirements specified herein. Unless otherwise specified (see 6.2), cast iron (except as noted in 3.5.9.4), malleable iron, and semi-steel shall not be used for any part of the anchor windlass.

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3.4 Environmental conditions. Unless otherwise specified (see 6.2), anchor windlasses shall be operable and have no loss of lubrication or liquids on ships encountering ±15 degrees of static list, ±5 degrees of static trim, ±45 degrees of dynamic roll (combatants), ±30 degrees of dynamic roll (non-combatants), and ±10 degrees of dynamic pitch. Anchor windlasses shall be operable in temperatures ranging from -40 to 140 °F (-40 to 60 °C) with relative humidity from 0 to 100 percent. Testing shall be conducted in accordance with MIL-STD-810 coastal/ocean, exposed (for weather deck machinery), and sheltered (for interior machinery) criteria (see 4.4).

3.4.1 Noise. Anchor windlass equipment shall not exceed MIL-STD-1474, grade D airborne sound level acceptance criteria and MIL-STD-740-2, type II structureborne noise level acceptance criteria.

3.4.2 Shock. All anchor windlass system equipment shall be in accordance with MIL-DTL-901, grade B shock requirements. In addition to the ship specification requirements, the portion of windlass equipment required for being towed by a similar ship in accordance with S9086-TW-STM-010/582 shall be in accordance with MIL-DTL-901, grade A shock requirements. This is to include the wildcat, spud, wildcat band brake and linkages, manual portions of the wildcat brake control, shaft, clutch, and any other equipment required for such towing. When specified (see 6.2), the entire anchor windlass system shall meet grade A shock requirements in accordance with MIL-DTL-901.

3.4.3 Vibration. All rotating machinery shall be balanced in accordance with MIL-STD-167-1 to minimize vibration. The windlass shall operate continuously and reliably during and after exposure to type I vibration in accordance with MIL-STD-167-1 and shall withstand internally excited vibration in accordance with MIL-STD-167-1, type II vibration.

3.4.4 Electromagnetic interference (EMI) suppression. All anchor windlass equipment shall be bonded and grounded in accordance with MIL-STD-464 and MIL-STD-1310 to minimize adverse electromagnetic effects and to protect personnel from electrical shock.

3.5 Design. Anchor windlasses shall consist of one wildcat, one capstan head, one electric motor drive, controls, bedplates, gearing, brakes, control stands, software, controller, variable-speed drive (VSD) (for type II windlass only), human machine interface (HMI) screen (for type II windlass only), and all other operating and control accessories suited to handling the weight of an anchor, as specified (see 6.2). Should a two-speed (type I) windlass be selected, the mechanical components of the system, figure 1 provides representative guidance for vertical, electro-mechanical anchor windlasses. The following general design considerations for both type I and type II shall be met:

a. The windlass shall meet strength, rigidity, and durability requirements specified herein and withstand the stresses encountered during windlass operation, as specified (see 6.2). The windlass shall support all inertial loads associated with starting and stopping of the chain or prime mover during all modes of operation. If required, the maximum size and weight shall not exceed that specified (see 6.2).

b. The design of keys, feathers, splines, nuts, screws, bolts, and collars shall include positive means to prevent their working loose. The use of set screws shall be avoided wherever possible. When set screws are used, means shall be provided to prevent their loosening and working out.

c. Each anchor windlass shall be capable of normal lowering and hoisting of the anchor at an average rate of at least 6 fathoms per minute from a depth of 60 fathoms. A low speed, not to exceed 1 fathom per minute, shall be provided for anchor housing, anchor alignment, and engaging and disengaging the wildcat brake. No specific hoist rate is required at depths greater than 60 fathoms.

d. The minimum wrap angle for the chain wrap on the wildcat shall be 170 degrees when accounting for the design of ship decks, hawsepipes, and chain lockers. When specified (see 6.2), design accounting for two-windlass installations shall include use of either windlass for either anchor while ensuring the minimum wrap angle is maintained.

e. Where shafts pass through a deck, provisions shall be made to obtain thorough watertightness. In the design of the windlass and its machinery, consideration shall be given to the deflection of decks of not more than 1 inch in either direction. Flexible couplings and slip joints, or other approved means, shall be provided as necessary to care for derangements and jamming of the driving mechanism that might otherwise be caused by the deflection of the decks.

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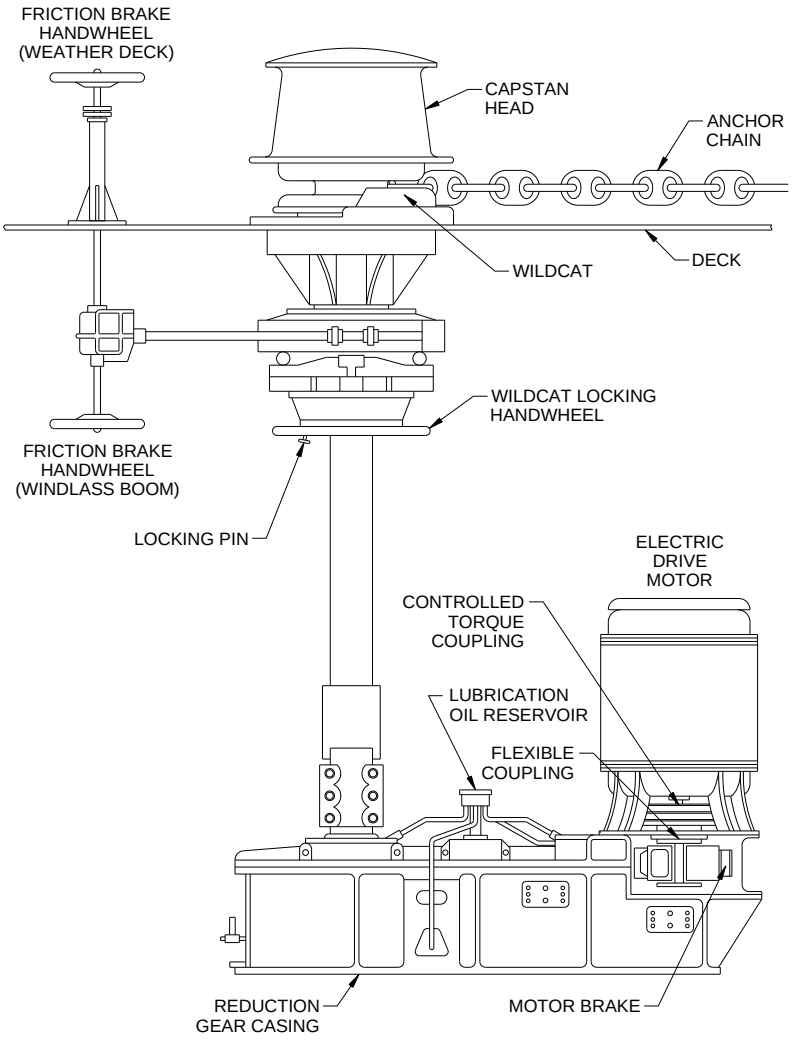


FIGURE 1. Typical type I vertical, electro-mechanical anchor windlass, representative only.

3.5.1 Welding. Welded fabrication may be used in lieu of castings. Fabrication, welding, and inspection shall be in accordance with S9074-AR-GIB-010/278 and AWS A2.4.

3.5.2 Material allowable stress. Unless otherwise specified herein, the material strength relationships specified in table I shall be used in design stress calculations.

TABLE I. Material allowable stress.

Design strength	Percent of the tensile yield point
Direct shear	60
Torsional shear	60
Compression (bearing)	160

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3.5.2.1 Normal operational design loads and allowable stresses. The rated normal working loads of heaving and hoisting the anchor and its chain from 60 fathoms at rated speed in normal operational conditions shall not produce combined stresses in any mechanical part that exceed 35 percent of the tensile yield point of the ductile materials used, unless approved by the Technical Authority (see 6.5). This includes all loads induced by the prime mover (limited by the VSD parameter settings for type II windlasses) and controller software during rated normal working load except as noted below:

a. Bolstering loads. For loads associated with normal bolstering (seating the anchor in its housing at low speed) and with operation in extreme environmental conditions (see 3.4), the combined stresses in any mechanical part shall not exceed 70 percent of the tensile yield point of the ductile materials used unless approved by the Technical Authority (see 6.5). This includes all functional and inertial loads induced by the prime mover (limited by the VSD parameter settings for type II windlasses) and controller software, hoisting and lowering the anchor to a 100-fathom water depth, anchoring in 70 knots of wind, 4 knots of current, and 240 feet (73 meters) of water. The calculated combined stresses in any mechanical part when bolstering at any speed above 1 fathom per minute shall not exceed 100 percent of the tensile yield point of the ductile material. The anchor windlass shall have automatic mechanical or electrical (non-procedural) methods in place to ensure that bolstering occurs at no greater than 1 fathom per minute (6 feet [1.83 meters] per minute).

b. Gearing. All gearing components, including planetary gearbox, keyway components, and splines shall be designed in accordance with ANSI/AGMA 945-A18, ANSI/AGMA 2101-D04, ANSI/AGMA 6001-E08, and ANSI/AGMA 6123-B06, depending on the style of gearbox used. Loads and service factors used in supporting American Gear Manufacturer Association (AGMA) calculations shall be traced to system operational loads, system requirements, and manufacturer data.

c. Component stress criteria. All anchor windlass original equipment manufacturer components shall be used within recommended manufacturer design load specifications. Load criteria considerations shall us the same factors as AGMA analysis (see 3.5.2.1.b).

3.5.2.2 Capstan load stress. Stresses shall not exceed 70 percent of the tensile yield point in any part of the equipment when the capstan head is subjected to a horizontal pull that is equal to the minimum breaking strength of the synthetic rope specified (see 6.2) and acting at mid-height of the capstan head and normal to the axis of the shaft. Capstan design load calculations shall be in accordance with MIL-DTL-17944.

3.5.3 Wildcat. The wildcat shall be designed to handle a flash butt welded stud-link chain that is in accordance with RR-C-271 and connecting links of the size specified (see 6.2). The wildcat shall be designed in accordance with ASTM F765, type IV and shall be a reversible, equally operable in either direction, vertical axis wildcat. Five-link and three-link sections (eight links total) of the specified size of chain and one detachable link shall be used for testing and ensuring the fit of the chain on the wildcat casting.

3.5.3.1 Drive. Each wildcat shall be arranged for positive drive through a driving head and locking head device located in the windlass room. The locking head shall be actuated by an operating mechanism of rugged construction that shall positively hold the drive head in the engaged or disengaged position. The entire operation of engaging or disengaging the wildcat drive shall be accomplished by one motion of a lever through an angle not exceeding 60 degrees. The operating gear and lever shall be conveniently located in the windlass machinery room. The wildcat and capstan head may turn together; however, the capstan shall be separately operable when the wildcat is stationary.

3.5.3.2 Clutch indication. A proximity switch shall be provided to sense the wildcat clutch position. Indicator lights shall be provided at the control stations to display the fully engaged and fully disengaged clutch positions. The clutch position shall be locally visible and properly lighted in the event that the clutch position indicators fail to operate. The system shall not operate if the clutch is partially engaged or disengaged.

3.5.3.3 Top. The top of the wildcat shall be designed with a step recess on the periphery and shall be fitted with a convex cover plate that has a rim that fits in the step recess. The cover plate shall be constructed of corrosion-resistant metal, bronze, or stainless steel; shall be secured with round head machine screws in accordance with FF-S-92; and shall be gasketed to prevent moisture from entering the interior of the hardware.

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3.5.3.4 Support. The wildcat shall be mounted on bearings and be supported such that the drive shafts are not subject to bending. The chain and rope pull operating stress and forces shall be completely taken by the deck bedplates.

3.5.3.5 Wildcat key fit. Wildcats shall be shrunk on their respective barrels or shafts and keyed with keys having a drive fit. Keyways shall be in accordance with ASME B17.1, class 1 fit or better. Keyway corners shall be designed with fillet radii to prevent local stress concentration by operating forces.

3.5.3.6 Speed limiting device. When specified (see 6.2) and deemed necessary, speed limiting devices shall be approved by the Technical Authority (see 6.5). The intention of speed limiting devices is not to be active in normal operation, but to be an additional safety measure during free fall operation.

3.5.3.7 Chain strippers. Chain strippers shall be provided for freeing the chain from the wildcat when heaving in or veering out.

3.5.4 Capstan heads. Capstan heads shall be of rugged construction and proportioned for the rope size specified (see 6.2) in accordance with 803-860303.

3.5.4.1 Top. The top of the capstan head shall be in accordance with 3.5.3.3.

3.5.4.2 Support. The capstan head shall be mounted on bearings and be supported so that the drive shafts are not subject to bending. The chain and rope pull operating stress and forces shall be completely taken by the deck bedplates.

3.5.4.3 Capstan head key fit. Capstan heads shall be shrunk on their respective barrels or shafts and keyed with keys having a drive fit in accordance with 3.5.3.5.

3.5.5 Major electrical components. Major electrical components for electro-mechanical anchor windlass systems include electric motor, controller (type I and II windlasses), VSD (type II windlass), encoder (type II windlass), and HMI (type II windlass).

3.5.5.1 Electric motor. The electric motor shall be squirrel cage induction or a permanent magnet variable-speed motor and shall be in accordance with MIL-DTL-17060 and the following:

a. Service: Navy type A.

b. Voltage: 440V three-phase.

c. Frequency: 0 to 60 hertz.

d. Windings: single.

e. Duty: varying. The motor shall be sized to ensure that an over-temperature trip does not occur during any planned evolution. Planned evolutions include all operations at rated load, operations in depths of water up to 100 fathoms, and all evolutions on the anchor nomograph. Six fathoms per minute is required for depths less than 60 fathoms, and any speed is acceptable when operating in depths greater than 60 fathoms. Bolstering shall be performed at 1 fathom per minute.

f. Enclosure: dripproof protected in accordance with MIL-STD-108.

g. Horsepower: as required to meet all load conditions.

h. Torque:

(1) Locked rotor torque: 275 percent full load torque, minimum.

(2) Pull-up torque: 300 percent full load torque, minimum.

(3) Breakdown torque: 275 percent full load torque, minimum.

i. Speed: 0 to 1800 revolutions per minute (RPM) (for type II windlass); 1800 RPM and 300 RPM if a two-speed motor is used (type I windlass).

j. Design: MIL-DTL-17060, design D.

k. Ambient temperature: 32 to 122 °F (0 to 50 °C).

l. Ambient humidity: 95 percent at 90 °F (32 °C).

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m. Bearing type: ball.

n. Insulation: class F, sealed (vacuum pressure impregnations).

o. Degree of balance: precision.

p. Mounting: the motor shall be designed for inverted mounting. The motor shall also retain bearing mounting requirements and drip proof requirements without modification.

q. Airborne and structureborne noise levels: in accordance with MIL-STD-1474, category D, and MIL-STD-740-2, type II, respectively.

r. Thermal protection: stator windings shall be provided with thermal sensors in accordance with MIL-DTL-17060. In a thermal overload condition, the anchor windlass shall operate at both high speed and low speed. High speed is defined as 6 fathoms per minute and low speed is defined as 1 fathom per minute.

3.5.5.2 Drive motor resolver. Type II windlass drive motors shall be provided with a suitable resolver meeting the following characteristics:

a. Integral to the motor frame.

b. Modular digital.

c. Temperature range: 32 to 311 °F (0 to 155 °C).

d. Input voltage 5.0 to 15.0 volts direct current.

e. Channels A, B (incremental), Z (marker).

f. Outputs designated as follows:

(1) Shield.

(2) Common.

(3) Output voltage: 4 to 15 volts.

g. Cannon plug in accordance with SAE AS50151.

3.5.5.3 Chain counter encoder. The anchor windlass shall be provided with a chain counter encoder meeting the following characteristics:

a. Integral to the anchor windlass system.

b. Modular digital.

c. Frequency response: 0 to 500 kilohertz.

d. Code format: SSI binary.

e. Supply voltage: 10 to 30 volts direct current.

f. IP67 rated.

g. Operating temperature: -40 to 212 °F (-40 to 100 °C).

h. Max shaft speed: 10,000 RPM (continuous).

i. Channels A, B (incremental), Z (marker).

j. Outputs designated as follows:

(1) Shield.

(2) Common.

(3) Output current: 60 milliamps per bit.

k. Cannon plug connector in accordance with SAE AS50151.

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3.5.5.4 Motor controller. Type I windlasses shall be provided with an electric motor controller in accordance with MIL-DTL-32613. A two-speed, alternating current (AC) magnetic, size 5 motor controller shall control the motor and electric brake. The directional contactors shall be mechanically and electrically interlocked so that "hoist" and "lower" cannot be activated simultaneously. The speed contactors shall be mechanically and electrically interlocked so that high and low speeds cannot be activated simultaneously. The motor controller shall include a "power on" indicator light, as well as emergency run, which shall bypass motor protections and reset pushbuttons. The motor controller shall prevent operation of drive machinery if the band brake is in the fully set position. The motor controller shall include an electronic overload that can be programmed to protect the motor. The motor controller shall include a three-position control station selector switch. The center position shall be off, the clockwise position shall be the windlass level control station, and the counterclockwise position shall be the motor level control station.

3.5.5.5 Programmable VSD. Type II windlasses shall be provided with VSD to control the motor and brake in accordance with MIL-PRF-32168, MIL-DTL-32613, and the following:

a. The drive shall be compatible with the motor.

b. The drive shall be provided with a lockable communication port to restrict physical access.

c. The drive shall be provided with a torque-proving circuit to verify motor load-holding capability prior to releasing the hoist machinery brake. The torque-proving circuit shall not permit the electric machinery brake to be electrically released until the hoist motor has developed sufficient torque to positively control the load.

d. The drive shall be provided with a torque-holding circuit such that the drive deceleration rate shall completely bring the motor rotation to zero speed while holding the load up to the locked rotor torque prior to de-energizing the electric machinery brake solenoid. The torque-holding circuit shall hold the motor at zero rotational speed to ensure that the machinery has stopped rotation before setting the electric brake. While the anchor is being housed in the bolster, the VSD shall provide sufficient power to prevent the load from back driving the motor until the electric brake is set.

e. The drive shall be programmed for a high speed of 6 fathoms per minute, a low speed of 1 fathom per minute, and a creep speed of less than 1 fathom per minute.

f. The drive shall be capable of being programmed to automatically reduce speed to 1 fathom per minute or lower prior to the anchor being housed in the bolster.

g. The drive shall utilize a closed loop vector control of the variable-speed motor.

h. The drive shall be microprocessor-controlled.

i. The drive system shall store drive programs and programmable parameters in removable non-volatile memory.

j. The drive shall include an operator interface that provides real-time display of drive status, faults, input and output state, and drive parameters. The operator interface shall allow adjustments to be made to programmable parameters of the drive and shall check the valid syntax and semantics of inputs, such as character set, length, numerical range, and acceptable values.

k. The drive shall display warning and fault indications for over temperature, under voltage, overcurrent, overvoltage, ground fault, communication loss, and loss of coolant if applicable.

l. Drive programmable features shall be capable of being password protected to prevent unauthorized modification.

m. The drive system shall operate with a supply power factor of 0.8 lagging to 0.95 leading. The power factor of the network bridge shall be 1.0 to 0.95 lagging for motoring and regenerating.

n. The drive system shall provide variable speeds utilizing a single winding motor. The drive shall be programmable to allow motor speeds of 0 to 1800 RPM. The speed holding accuracy shall be 0.02 percent of full speed from 0 to 100 percent motor load. The drive shall not require an external braking resistor to maintain speed regulation requirements.

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o. The drive shall incorporate parameter file and hardware integrity checks that shall prevent operation of the drive using incompatible software. The software and hardware checks shall verify that drive software matches the drive configuration at startup. In the event a software program does not match the drive configuration, the incompatible software shall not be executed, and a fault message identifying the incompatible software shall be accessed through the diagnostic terminal.

p. The drive system shall provide diagnostic capabilities to allow faults to be isolated to the lowest replaceable unit or module. Diagnostics shall be of power up and on-line type. The drive shall display the diagnostic result, the status of selectable addresses, and the list of suspected modules that could cause the failure. The elimination of suspected failed modules shall be a maximum of three of the most likely modules that could cause that particular failure.

q. The drive shall be provided with an Ethernet port. This port shall allow monitoring, troubleshooting, adjusting of programmable parameters, and uploading or downloading of drive software via a portable computer utilizing the VSD system software.

r. All drive statuses, diagnostic and fault information, troubleshooting features, and programmable parameter adjustments shall be accessible through the machinery room diagnostic terminal.

s. The drive shall be capable of operating the electric machinery brake specified. The drive shall have a dedicated brake output that is hardwired to the coil of the electric machinery brake.

t. The drive shall be capable of operating the anchor windlass in an emergency mode in the event of a programmable controller failure or overload, which shall bypass the programmed controls.

u. All fuses shall have blown fuse indicator lights.

v. The drive shall be dripproof protected in accordance with MIL-STD-108.

w. The drive shall prevent operation of drive machinery if the band brake is in the set condition.

x. Drive programmable features shall be capable of being password protected to prevent unauthorized modification.

3.5.5.6 Programmable logic controller (PLC). The PLC shall perform all anchor evolutions in accordance with MIL-DTL-32006, MIL-DTL-32613, and the following:

a. The PLC shall not replace any functionality of the safety circuit.

b. The PLC shall employ anti-tamper technologies to identify physical tampering or alteration.

c. PLC programmable features shall be capable of being password protected to prevent unauthorized modification.

d. The PLC shall incorporate integrity checks that shall prevent operation of the PLC using an incompatible project file. The project file-hardware checks shall verify that the project file matches the PLC configuration. In the event a project file does not match the PLC configuration, the project file shall not be executed at startup, and a fault message shall be accessible through the diagnostic terminal.

e. The PLC shall contain removable non-volatile memory for storage of the usage data, project file, and parameters.

f. The PLC shall be supported by an HMI. All PLC statuses, diagnostic and fault information, troubleshooting features, and programmable parameter adjustments shall be accessible through the HMI.

g. All anchor switches, relays, and sensors shall be configured to provide individual inputs to the PLC.

h. The PLC shall provide system diagnostics to isolate faults or failures down to the lowest replaceable module level. Fault status outputs from devices forming part of the system shall be configured as individual inputs to the PLC. Additionally, the PLC processor, communications modules, and input/output (I/O) modules shall provide a self-diagnostic capability. Diagnostics shall be of the power up and continuous type. Failure of the processor, any communication device, I/O module, or communication network shall result in failure of the PLC to provide or maintain an output bit labeled “PLC Ready”. “PLC Ready” is an indication of the system health of the logic controller and its network and shall be an input to the PLC safety circuit.

i. The PLC shall be dripproof protected in accordance with MIL-STD-108.

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3.5.5.7 Touchscreen HMI. Type II windlasses shall be provided with an HMI for the primary operator interface to select and display all anchor system status, configuration, and diagnostic information. The HMI as provided as part of the anchor windlass system shall conform in accordance with MIL-DTL-32613 and the following:

a. The HMI shall contain non-volatile flash memory for the storage of the HMI program.

b. The HMI shall interface with the anchor windlass controller via an open architecture network communication protocol.

c. The HMI shall be provided with a lockable hinged or removable cover to prevent inadvertent display screen contact by personnel or objects. The cover shall not prevent personnel from reading the display screen when positioned to protect the screen.

d. All touchscreen HMIs shall display the following information:

(1) Current state of the VSD.

(2) Current state of the PLC.

(3) Current state of each switch and relay.

e. The HMI shall incorporate self-diagnostics. HMI fault status and diagnostic information shall be accessible through the HMI display.

f. The HMI shall have a reset button that reboots the HMI when depressed.

g. The HMI shall include a clutch engage mode to allow engagement and disengagement of the clutch at a speed of less than 1 fathom per minute.

3.5.5.8 Motor couplings. To prevent the anchor and chain from overhauling the windlass, an electric brake shall be attached to the speed reducer high speed pinion shaft. This design allows the brake to stop the windlass upon loss of power to the electric motor or when power is not applied. This is the preferred design, as failure of the flexible, all-steel coupling does not uncontrollably drop the load if the coupling fails. As an alternative, the hoist machinery electric motor can be connected to the speed reducer through a flexible all-steel coupling. The coupling shall be considered fail safe if the electric brake is not attached to the speed reducer high speed pinion shaft. The use of a friction clutch or other non-positive type of drive shall be prohibited. Grease lubrication shall be provided by high-pressure grease fittings and vents. Replacement of coupling wearable components shall not require electric motor removal or re-alignment.

3.5.5.9 Vertical shafts. Slip joints or couplings shall be provided for vertical drive shafts where deflection or warping of decks may cause misalignment of the shafts.

3.5.5.10 Electrical equipment interface. All major electrical equipment shall operate from a 440 volts alternating current (VAC), 60 hertz, three-phase high resistance ground, type I power system having steady-state and transient characteristics in accordance with MIL-STD-1399-300-1 and MIL-DTL-917. All other electrical equipment, including those utilizing 115 VAC single- or three-phase power, shall be in accordance with the electrical interface requirements set forth in MIL-STD-1399-300-1.

3.5.5.11 VSD disconnect switch. Type II windlasses shall be provided with a 600 VAC, three-pole, single-pole disconnect switch in accordance with MIL-DTL-17361. The disconnect switch shall remove 440 VAC three-phase supply power from the VSD and shall be sized for the maximum control system loading requirements. The disconnect switch shall be mounted so that it is accessible external to the controller enclosure. The disconnect switch shall be dripproof protected in accordance with MIL-STD-108.

3.5.6 Control stations. Controls shall be arranged for convenient and safe operation of the equipment. Controls shall remain in the position set by the operator. Controls shall be as follows:

a. Control stand windlass level: controls and indicators.

b. Control station motor level: controls and indicators.

c. Handwheel windlass and motor levels: operation of the associated wildcat brake (may be directly linked).

d. Remote controls such as at the bridge, if necessary and so specified (see 6.2), shall be approved by the Technical Authority (see 6.5).

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3.5.6.1 Control functions. At a minimum, the control functions shall include:

a. A vertically-mounted master switch that provides direction (hoist/lower) and speed selection and incorporates a dead man feature that returns to neutral when no force is applied.

b. A white indicator light for power on.

c. An emergency stop pushbutton switch to remove power from the motor.

d. White, fully engaged and fully disengaged indicator lights for the clutch.

e. A lamp test functionality for all indicator lights.

f. A lamp dimmer switch for all indicator lights.

g. A red overload protection indicator light.

h. A white station in control indicator light.

i. White indicator lights for normal or emergency mode of operation.

j. A red variable-speed fault indicator light (type II windlass).

k. A red PLC fault indicator light.

l. A chain counter display indicating length of chain payed out.

3.5.6.2 Windlass level controls. The windlass level control design shall provide a clear view of the operation of the associated windlass.

3.5.6.3 Motor level controls. The motor level shall contain a controller in accordance with MIL-DTL-32613. The motor level shall also include a handwheel in close proximity to the associated wildcat brake. These stations shall be used during maintenance and outside of normal operations. There shall be a means to turn power on and off and to select the control station to be used on the motor controller or VSD rather than at each control station. Each control station shall be of watertight construction. Watertight designation shall be in accordance with MIL-DTL-2036. When power is applied to the motor controller, the station that the user would like to operate shall be able to be selected. The station in control shall have a lit indicator light on its control station, while the non-selected station shall have a non-lit indicator light.

3.5.6.3.1 Emergency stop pushbutton. The pushbutton shall be red in color and surrounded by a protective bezel to prevent inadvertent operation. The emergency stop pushbutton shall be push to open and pull to close. The pushbutton shall be a mushroom-type switch. The pushbutton shall be located away from other indicators and operators such that inadvertent operation is avoided. The emergency pushbutton shall cause loss of all power to the drivetrain and cause all action to stop immediately. Reset of the emergency pushbutton shall not cause automatic restart. The emergency stop pushbuttons shall be hard wired to the motor controller or VSD and shall not rely on the PLC software for operation. The emergency stop pushbutton shall cause the PLC software to halt until the emergency stop pushbutton is reset and a new movement command is received.

3.5.6.3.2 Lamp dimmer. The dimmer shall control all indicators, including power indication and overload indication. In the full “on” condition, the lamps and indicators shall be visible in daylight. Near the “off” condition, lamps and indicators shall be capable of night-time operation.

3.5.6.4 Water protection. The motor controller, VSD (type II windlass), and any other electrical or electronic items shall be in drip-proof enclosures. The control stands shall be watertight. Drip-proof and watertight protection designations shall be in accordance with MIL-DTL-2036.

3.5.6.5 Cabling. Electrical cabling furnished with equipment should be selected using MIL-HDBK-299 as guidance.

3.5.6.6 Cable connectors. All electrical switches, connections, and lighting fixtures shall be protected by watertight fittings. Cable connectors shall be in accordance with SAE AS50151.

3.5.6.7 Shielding. Shielding on wires and cable shall be secured to prevent contacting or shorting of exposed current-carrying parts where necessary. To prevent fraying, the shielding shall have secured or braided ends.

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3.5.6.8 Mechanical aspects. The control mechanical design shall consist of joints, shafts, and gears mounted on antifriction bearings and arranged for minimum lost motion. Bearings and gears shall be completely enclosed. Bearings shall be spaced and located to prevent whipping and excessive deflection of control shafting. Steel tubing may be used for shafting where practicable to save weight and secure torsional stiffness. Universal joints or splined slip joints shall be provided where necessary to allow for deflection and working of the ship structure. Universal joints shall be secured to control shafts with keys locked by self-locking set screws or by other means. Stuffing boxes shall be furnished where control shafting passes through watertight decks or bulkheads. Roll pins, tapered pins, and bolts shall not be used in joints of the control shafting.

3.5.7 Bedplates. Bedplates shall be of cast or welded steel construction so as to maintain alignment of all parts, to transmit external forces to the ship's structure without objectionable deflection, and to resist the shock forces specified in 3.5.2. Bedplates to be installed on weather decks or in the anchor windlass room shall be watertight with no pockets or hollows in which water or debris may collect. The underside of bedplates shall be finished to a plane surface for watertight mounting on foundations.

3.5.7.1 Wildcat bedplates. Each wildcat shall be mounted on a bedplate, which may consist of one piece or two pieces bolted together. When specified (see 6.2), the bedplate shall contain chain fairleads from the hawsepipe and the chain pipe to the wildcat. Fairleads shall be formed to an easy radii and shall take up not less than three complete links of the chain. The height of fairleads and chain pipe deck bolsters shall be such as to deliver the chain to the horizontal center line of the wildcat between whelps. The location of the chain pipe deck bolsters shall allow for handling the chain in and out of the wildcat. The angle of rake and the details for connecting the chain pipes to the chain pipe deck bolsters shall be in accordance with ISO 4568 or as specified (see 6.2). The radii of flared mouth of the chain pipe deck bolsters leading to the wildcats shall be such that the chain binding is minimized. When specified (see 6.2), chain pipe deck bolsters shall be fitted with portable watertight metal bucklers fastened by means of hinged drop bolts with non-detachable bronze nuts.

3.5.7.2 Power unit bedplates. Each power unit shall consist of an electric motor, reduction gearing (if required), control equipment, and related accessories. Each power unit shall be mounted on a bedplate in the motor room. Bearing pads for units mounted on bedplates shall be machined to flat surfaces, and fitted bolts shall be used to secure the units to bedplates. Tap bolts shall not be used.

3.5.8 Gears. Gears shall be designed in accordance with ANSI/AGMA 945-A18, ANSI/AGMA 2101-D04, ANSI/AGMA 6001-E08, and ANSI/AGMA 6123-B06, depending on the style of gearbox used, to transmit and withstand the full torque of the windlass.

3.5.8.1 Main gear reducer. High-speed gear reduction between the electric motor and the windlass shafting shall be mounted on antifriction bearings, fully enclosed, and positively lubricated. A gearbox shall be provided between the electric drive motor assembly and the wildcat low-speed gearbox. The low-speed gearing shall be totally enclosed with inspection covers. Low-speed gearing lubrication shall be swabbed grease via inspection cover access. The planetary gears shall be lubricated by an oil bath in an oil-tight case. The gear cases shall be constructed from non-brittle ferrous material. The interior of the housing shall be coated with a permanent, epoxy-based primer in accordance with MIL-DTL-24441. All gear cases shall be designed so that, in all planned evolutions, including all evolutions outlined on the nomograph as well as recovery of the anchor in up to 100 fathoms of depth, the gear case oil temperature does not exceed manufacturer limits. The gear cases shall have an attached thermometer to monitor oil temperature and a sight glass or dipstick to monitor oil level during operation. These indicators shall be marked with acceptable ranges. A gear oil valve type sampling port shall be provided. Gear case joints shall be oil-tight. An air vent that is baffled to prevent the loss of oil under operating conditions and of a type that excludes foreign matter shall be provided for each gear case. Lifting eyebolts shall be installed as necessary on gear cases. Access to the gear mesh points for inspection shall be as easy as practicable. Where the design of the gear boxes precludes inspection without disassembly, provision shall be made for the inspection of all internal gears and the repair and replacement of the gear box assembly. The design shall provide adequate accessibility to enable the disassembly, inspection, and repairs of all gear box assembly components.

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3.5.8.2 Gear case. Gearing shall be fully enclosed in an oil-tight case arranged for the ready removal of gears and bearings. Cases shall be provided with bolt- or screw-fastened oil-tight inspection covers, fills, and drain openings equipped with oil sample valves. Oil-tight inspection covers shall be provided in sufficient number to allow visual examination of the teeth on all gears, permit cleaning of the gear case bottom, and allow inspection of all oil-lubricating pump strainers. Gear cases of 5 gallon (19-liter) capacity and larger shall be fitted with quick-disconnect suction and return lines to permit installation and fluid maintenance.

3.5.9 Brakes. Hand and electric brakes shall be provided. The wildcat shall be fitted with an externally contracting, hand-operated band brake located below the weather deck and controlled by a screw-type handwheel. Manual operation shall be from a horizontal handwheel located in the anchor handling area(s) near the control station(s). Each brake handwheel control station shall have a mechanical pointer that indicates engagement or disengagement.

3.5.9.1 Band brake capability. The brake shall stop and hold the anchor and chain in no more than the below listed rates per depth:

a. 15 fathoms out: stop within 2 fathoms.

b. 30 fathoms out: stop within 2 fathoms.

c. 45 fathoms out: stop within 3 fathoms.

d. 60 fathoms out: stop within 4 fathoms.

3.5.9.2 Brake bands design. The wildcat brake bands shall be lined with non-asbestos, non-burn brake band lining in accordance with A-A-52522, type I. Means shall be provided for lifting and keeping brake bands free of their drums when brakes are released. Necessary supports and guides that ensure smooth and easy operation of brake bands and operating mechanisms shall be furnished. The band brake shall meet the following requirements:

a. The ship shall be provided a band brake contact inspection method with adequate access to visually confirm full contact around the brake drum and minimum thickness requirements to verify the status of the band components.

b. The anchor windlass mechanical brake shall be sized such that thermal loads associated with acceptance and periodic testing accomplished in rapid succession (in approximately 1 minute between each test) do not exceed the thermal capacity or pressure capacity of brake components as provided by the brake pad manufacturer. Band brake temperature criteria shall not exceed the brake pad manufacturer criteria for estimated fade temperature.

c. The brake band design shall be able to withstand the total free fall arresting energy of the anchor, accounting for anchor and chain weight, at 60 fathoms. The effective braking area shall be such that at a velocity of 30 feet per second, and stopping within the 5 fathoms requirement, the heat transfer rate per unit area of the brake pad shall not exceed the maximum brake pad manufacturer criteria for continuous use. Brake band thermal capacity shall be calculated assuming five complete stops per minute using the following formula:

$$TC = ET * n/550$$

"ET" represents the total kinetic energy of all moving elements to be absorbed by the brake, and "n" represents the required number of stops per minute (five). The brake shall be capable of fully absorbing and dissipating this energy at a normal ambient temperature of 72 °F. The brake band pad material shall have friction characteristics of SAE J866 code GG or better and a wear rate of no more than 0.005 $inch^3$/hp-hr, as determined by testing in accordance with SAE J661.

d. The drum temperature shall not exceed 80 percent of the brake pad material temperature requirement from the brake pad manufacturer when conducting the 15, 30, 45, and 60 fathom performance requirement tests in rapid succession (in approximately 1 minute between each test).

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3.5.9.3 Hand brake power assist. Anchor windlasses that handle anchors weighing 30,000 pounds and over or those that are designed to handle anchor chains of 3½-inch wire diameter and larger shall be provided with power-assist control as well as manual screw-type handwheel control for applying the wildcat brake. Powered assist shall be provided to enable a 5 percent female in accordance with MIL-STD-1472 to operate the system during all conditions for life of ship. The power-assist control shall be located on the manual handwheel control. Setting of the brake by power assist, when utilized, shall be accomplished by a partial turn of the handwheel that is used to provide manual operation. Means shall be provided for adjusting the brake application force under power-assisted operation so as to avoid a slamming on effect of the brakes. When selected by the operator, each wildcat brake control, power-assist, or manual screw-type handwheel shall apply the wildcat brake in all modes of windlass operation without requiring any change or adjustment to windlass components. If powered assist is not utilized, calculations showing that powered assist is not needed for operation by a 5 percent female in accordance with MIL-STD-1472 over the life of ship shall be demonstrated.

3.5.9.4 Band brake mechanical components. All handwheels, levers, and associated members shall be designed in accordance with the following:

a. A two-man tight operating force of 350 pounds shall be allowed, applied to the rim or handle when the mechanism is driven against its stopped or extreme position, without exceeding 70 percent of ductile yield of all band brake mechanical components.

b. The effort required at the rim of the brake handwheel to set the brake shall not exceed 100 pounds of force.

c. Joints shall be such that the mating components are secured with two keys that are 90 degrees apart. The keys shall be positively secured with self-locking set screws or by other means.

d. Stresses shall not exceed 25 percent of the tensile yield point in any part of wildcat brakes or their operating mechanisms and mountings under any of the operation or test conditions specified herein.

e. The deck-band brake-control handwheel shall be horizontally mounted on a pedestal and located with full regard to protect operating personnel from the whip of the bitter end of the chain. The height above deck shall be at the average user's waist height, and dimensions of the handwheel shall be in accordance with MIL-STD-1472. The handwheel shall turn clockwise to set the brake and shall be marked to indicate the direction of rotation for setting and releasing the brake. The number of turns from set position to complete off position shall not exceed three turns. The brake operating gear and mechanism shall be non-overhauling and self-locking. An indicator showing an "off" and "on" position shall be provided at each control station. The indicator shall not restrict motion of the handwheel when it is in the fully "on" or fully "off" position.

f. In a two-man tight condition, the brake shall not slip prior to the time needed to enable the installation of the chain stoppers.

3.5.9.5 Electric/fail-safe brake. To prevent the anchor and chain from overhauling the anchor windlass, the brakes shall be arranged to set upon loss of power or set when power is not applied to the electric motor. Electric brakes shall be disc type with brake characteristics in accordance with MIL-DTL-16392 and the following:

a. Electric brakes shall have sufficient capacity to stop and hold 150 percent of the weight of the anchor and 80 fathoms of chain while being lowered at full speed under control of the drive assembly.

b. Supply voltage: 440 VAC, three-phase, 60 hertz.

c. Mounting: horizontal speed reducer, high speed pinion shaft, flanged.

d. Enclosure: Dripproof protected in accordance with MIL-STD-108.

e. Brake coil: class H insulation.

f. Rotating brake discs shall be provided with heavy-duty lining bonded to a bronze carrier ring. Asbestos shall not be used.

g. Stationary discs and pressure plate shall be nodular or ductile iron in accordance with MIL-C-24707/5.

h. Brake overheating shall not cause damage other than brake lining wear.

i. Brake-mounting surfaces shall retain their flatness within the original tolerances regardless of overheating and torque.

j. The brake shall not be vulnerable to internal corrosion and shall never require internal anti-corrosion preservatives that could interfere with performance.

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k. The brake shall be designed to easily permit brake lining wear checks.

l. The brake shall be self-adjusting.

m. The brake shall be easily cleanable so that dust resulting from brake wear shall not interfere with brake operation or reduce brake life when the brake is cleaned regularly.

n. The brake shall include a manual brake release mechanism. The manual brake release shall be operable by one person without the use of tools and without disassembly of any portion of the hoist machinery. Operation of the brake release shall be through a single lever or handle. The brake release shall not be detented, shall not latch in any position, and shall be spring returned. The brake shall re-engage when manual input to the brake release is discontinued. The brake manual release shall permit manual control of the anchor windlass clutch assembly from engaged to disengaged position.

o. Supports and covers shall ensure that oil or grease does not leak or drip onto brake surfaces.

3.5.9.6 Wildcat movement. Brakes for all windlasses shall be designed to meet all requirements specified herein regardless of the wildcat's directional movement.

3.5.9.7 Manual brake effort. The effort required at the rim of the brake handwheel to set the brake shall not exceed 100 pounds of force under any condition.

3.5.9.8 Brake handwheel. The deck band brake control handwheel shall be horizontally mounted on a pedestal and located with full regard to protect operating personnel from the whip of the bitter end of the chain. The height above deck and dimensions of the handwheel shall be in accordance with MIL-STD-1472. The handwheel shall turn clockwise to set the brake and shall be marked to indicate the direction of rotation for setting and releasing the brake. The number of turns from the set position to the complete "off" position shall not exceed three turns. The brake operating gear and mechanism shall be non-overhauling and self-locking. An indicator showing an "off" and "on" position shall be provided at each control station.

3.5.9.9 Brake control rod joints. Joints shall be such that the mating components are secured with two keys that are 90 degrees apart. The keys shall be secured positively with self-locking set screws or by other means.

3.5.10 Software. Any and all software, firmware, and programs necessary to configure, operate, monitor, maintain, and program the system, and any programmable devices contained therein, shall be as specified (see 6.2 and 6.4).

3.5.11 Lubrication. Means for lubrication of all working parts from readily accessible locations shall be provided. This equipment is known to be inactive for extended periods of time; therefore, splash and transfer lubrication methods shall not be used. If a power-driven oil pump is provided, it shall be furnished for spray-type lubrication of gears and bearings of the main gear drive between the wildcat and capstan shafts. If spray-type lubrication systems are provided, they shall be active when equipment controls are powered. The oil used shall be as determined by the windlass manufacturer. Pressure grease lubrication shall be provided for all other bearings. All bearings in gear housings not completely lubricated by oil in the housing shall be fitted with grease fittings for pressure grease lubrication in accordance with SAE AS35411. Large bearings, such as wildcat and capstan head bearings, shall be provided with several lubricant fittings, but not less than three fittings, equally spaced on the periphery of the bearing. Each lubricating fitting shall be installed at a safe distance from rotating components of the anchor windlass and, if possible, at one common place.

3.6 Reliability and maintainability. Units shall meet or exceed the following requirements:

a. Mean time between (mission critical) failure: 1,000 operational cycles as defined in 4.6.5.1, or as specified (see 6.2).

b. Unit design life: 30 years or as specified (see 6.2).

c. Mean time to repair: 6.0 hours or as specified (see 6.2).

d. Minimum time between overhauls: 250 operational cycles or as specified (see 6.2).

e. Statistical confidence of 80 percent or as specified (see 6.2).

3.7 Performance characteristics.

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3.7.1 Anchor hoisting. Without exceeding the full-load rating of the motor, the windlass shall hoist an anchor and chain of the specified weight and size (see 6.2) from a depth of 60 fathoms at an average rate of at least 6 fathoms per minute and from a depth of 100 fathoms up to 60 fathoms at no specified speed.

3.7.2 Capstan. Capstan heads shall develop a rope speed of at least 125 feet (38 meters) per minute with the specified size of rope (see 6.2) at no load condition. The capstan shall also withstand a static load equal to the minimum breaking strength of the specified size of rope using the head as a bitt at the mid-height of the head and normal to its axis. The capstan shall be able to pull rated load at rated speed. Under either load, the combined stresses in any part of the equipment shall not exceed specified allowable stresses (see 3.5.2).

3.7.3 Brake performance. Brakes shall be of ample size and strength to permit performance of specified brake tests without combined stresses in any part of the brake or operating mechanism exceeding the allowable stresses (see 3.5.2).

3.7.3.1 Hand brake. The hand brake shall meet the requirements of 3.5.9.1.

3.7.3.2 Electric/fail-safe brake. Electric brakes shall meet the performance requirements of 3.5.9.5.

3.7.4 Chain handling. The windlass system shall be capable of executing towing and mooring procedures in accordance with S9086-TW-STM-010/582.

3.8 Human factors. The anchor windlass system design shall accommodate normal operation by personnel in the 5th through 95th percentile in accordance with MIL-STD-1472.

3.9 Safety.

3.9.1 Operational safety. The design shall include guards and protective items to minimize the potential for personnel injury in accordance with MIL-STD-882.

3.9.2 Loss of power. Manual operation of the windlass system for lowering the anchor shall be possible in the event of electrical power loss. Failure of the power source of any power-operated drive mechanism or control shall not result in damage to the anchor windlass equipment, jeopardize the safety of personnel, or result in uncontrolled movement of the equipment and load. In the event a power loss occurs during operation, the clutch shall be able to move from the “engaged” to the “disengaged” position.

3.9.3 Control system safety features. Safety feature components and controls shall not be exposed to damage under normal operating conditions or be exposed to encourage possible tampering by operating personnel where such tampering could cause the safety feature to become disabled.

3.10 Dimensions and weight. The size of the chain, anchor, and rope shall be as specified (see 6.2).

3.11 Paints and color. Coatings shall be applied shortly after manufacture and cleaning. Metal to be coated shall have surfaces prepared to “near-white” metal in accordance with SSPC-SP 10 (WAB)/NACE No. 2. The abrasive blast surface profile shall be 2 to 4 mils.

3.11.1 Painted surfaces. Anchor windlass external surfaces shall be coated with two coats of epoxy in accordance with MIL-PRF-23236, type VI, class 5, and two topcoats of silicone alkyd in accordance with MIL-PRF-24635, type III, class 2, grade C. The silicone alkyd shall be interior haze gray (26270 of SAE AMS-STD-595). Unless otherwise specified (see 6.2), coatings shall be applied in accordance with the coating manufacturer’s product data sheet.

3.11.2 Unpainted surfaces. Bearing and machined surfaces or interior “oil-wetted” surfaces shall not be painted. Corrosion-resistant materials shall not be painted.

3.12 Identification plates. Identification plates and other designated marking plates shall be in accordance with MIL-P-15024/5. All plates, including operating and lubrication charts exposed to the weather, shall be fabricated from corrosion-resistant steel and shall be type A, B, or C. Charts shall be mounted by fully bedding with adhesive and attached with corrosion-resistant steel fasteners. Identification and marking shall be in accordance with MIL-STD-130.

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3.13 Workmanship. The windlasses and components shall be free from imperfections that might impair their serviceability or present hazards to operating personnel. The external finish shall be clean, smooth, and provide complete coverage. There shall be no signs of blistering, corrosion, discoloration, scrapes, gouges, or voids in the finish. Welds shall be free of harmful defects, such as cracks, undercuts, voids, or gaps. There shall be no sharp or ragged edges, corners, burrs, or protrusions that might create a safety hazard to personnel. There shall be no dents or loose, damaged, bent, or broken parts or fasteners. There shall be no defects that might reduce the reliable operational capability or life of the equipment. There shall be no signs of fluid leakage. Piping and cabling shall be supported by clamps or hangers. Cabling and wiring insulation shall not be nicked, gouged, cut, or otherwise damaged. Safety guards, covers, shields, and warning plates shall be appropriately installed and marked.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on electro-mechanical anchor windlasses when a first article sample is required (see 3.1). This inspection shall include the tests specified in table II and the requirements of ISO 4568.

4.3 Conformance inspection. Conformance inspection shall include the tests specified in table II.

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TABLE II. First article and conformance inspection and testing.

Characteristic		Requirement	First article	Conformance
Materials		3.2, 3.3	4.5	4.5
Environmental conditions		3.4	4.4, 4.6.1	--
Noise		3.4.1	4.6.1.3	--
Shock		3.4.2	4.6.1.4	--
Vibration		3.4.3	4.6.1.5	--
EMI		3.4.4	4.6.1.6	--
High temperature		3.4	4.6.1.7	--
Low temperature		3.4	4.6.1.8	--
Humidity		3.4	4.6.1.9	--
Design		3.5 through 3.5.5.11	4.5	4.5
Shop	Pre-checks/no load	3.8, 3.9	4.6.2.1	4.6.2.1
	Normal operation	3.7.1	4.6.2.2	4.6.2.2
	Wildcat hand brake	3.5.9.1, 3.7.3.1	4.6.2.3	4.6.2.3
	Electric/fail safe brake	3.5.9.5, 3.7.3.2	4.6.2.4	4.6.2.4
	Wildcat chain	3.5.3	4.6.2.5	--
Dock	Pre-checks/no load	3.8, 3.9	4.6.3.1	4.6.3.1
	Normal operation	3.7.1	4.6.3.2	4.6.3.2
	Wildcat hand brake	3.5.9.1, 3.7.3.1	4.6.3.3	4.6.3.3
	Electric/fail safe brake	3.5.9.5, 3.7.3.2	4.6.3.4	4.6.3.4
	Capstan loads	3.7.2	4.6.3.6	--
	Safety	3.9 through 3.9.3	4.6.3.2	--
Deep sea	Hoisting	3.7.1	4.6.4.1	4.6.4.1
	Wildcat hand brake	3.5.9.1, 3.7.3.1	4.6.4.2	4.6.4.2
	Electric/fail safe brake	3.5.9.5, 3.7.3.2	4.6.4.3	4.6.4.3
	Stowage	3.5.5.1.e	4.6.3.4	--
Chain handling		3.7.4	4.6.4.4	4.6.4.4
Reliability and maintainability		3.6	4.6.5	--
Software		3.5.10	4.6.6	4.6.6
Cybersecurity		3.5.5.4; 3.5.5.5; 3.5.5.5.b, .e, .f, .g, .h, .i, .j, .l, .o, .p, .r, and .x; 3.5.5.6.a through .h; 6.3	4.6.7	--
Human factors, dimensions, paints, identification plates, workmanship		3.8, 3.10, 3.11, 3.12, 3.13	4.5	--

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4.4 Test conditions. For the tests specified in 4.6, the equipment is not required to operate under full-load conditions for more than 1 hour in a 4-hour period. Unless otherwise specified (see 6.2), environmental conditions shall meet MIL-STD-810, coastal/ocean, exposed (for weather deck machinery), and sheltered (for interior machinery) criteria.

4.5 Examination. Anchor windlasses shall be visually examined for compliance with the requirements specified in 3.5 through 3.5.8.2, 3.8, and 3.10 through 3.13. This element of inspection shall encompass all visual examinations, dimensional measurements, and calculation verifications contained in those requirements.

4.6 Tests.

4.6.1 Environmental tests.

4.6.1.1 Equipment to be tested. The following equipment, which may experience degradation or operational impacts due to environmental effects, shall be tested:

a. Motor, to include motor protective devices, and, for type II units, VSD operation and settings.

b. Control stations, to include use of manual handwheels.

c. Controllers.

d. Controls, to include emergency stop pushbutton and reset aspects.

e. Chain counter.

f. Other electric cabinets and relays.

4.6.1.2 Readings to be taken. The following readings shall be taken during no-load and load tests:

a. RPM of wildcat shaft.

b. Motor voltage, amperes, frequency, temperature, RPM, vibration, and unusual noise shall be noted.

c. Oil pressure and temperature. Fluid type and allowable temperature shall be recorded for comparison.

d. Time (start, stop, and duration) and number of fathoms completed during that time.

e. Hoisting speed.

f. Performance as defined herein of the hand brake.

g. Maximum recorded and allowable temperatures of gearbox input bearing and wildcat bearing before, during, and after tests.

h. Torque of the system shall be measured to capture total torque of the system (during shop testing only).

i. Temperature of brake pads during testing. Infrared detection may be used. Maximum value shall be noted.

j. All faults seen during testing from the VSD (type II windlass only).

k. Any change in coupling status (i.e., wear or temperature).

l. Electric brake wear and installation measurements.

m. Electric brake temperature before and after loaded operations.

n. Any duty cycle time outs.

o. Infrared camera observations during testing.

4.6.1.3 Noise testing. The airborne noise shall be measured in accordance with the instrumentation and procedures of MIL-STD-1474 with the exception that noise levels at the measurement positions shall be averaged. Motors shall be tested while operating at no load with rated voltage and frequency applied. Structureborne noise shall be tested in accordance with MIL-STD-740-2.

4.6.1.4 Shock test. Components shall be tested for high-impact shock in accordance with MIL-DTL-901.

4.6.1.5 Vibration test. The anchor windlass shall be tested in accordance with MIL-STD-167-1, type I, ranging from 4 hertz (or lowest attainable frequency) through 50 hertz, and type II.

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4.6.1.6 EMI test. The equipment and associated electronics shall be tested for EMI in accordance with MIL-STD-461.

4.6.1.7 High temperature test. The equipment shall be subjected to a high temperature test in accordance with MIL-STD-810, method 501.4, procedure II. The cyclic temperature test shall be performed.

4.6.1.8 Low temperature test. The equipment shall be subjected to a low temperature test in accordance with MIL-STD-810, method 502.4, procedure I.

4.6.1.9 Humidity test. The equipment shall be subjected to a humidity test in accordance with MIL-STD-810, method 507.6.

4.6.2 Operational shop tests.

4.6.2.1 No-load test. With the wildcat engaged, each anchor windlass shall operate continuously with no external loading for at least 30 minutes at full speed. Direction shall be alternated several times such that each direction receives 50 percent of the total time.

4.6.2.2 Normal operation test. The anchor windlass shall be subjected to a test that simulates normal operation, up to full rated capacity, to determine conformance to the requirements of 3.9. A brake system may be used to provide the necessary torque resistance equal to 30 fathoms of chain and anchor. The speed shall be at an average of 6 fathoms per minute without exceeding the full-load rating of the motor or controller. The time duration shall be a minimum of 10 minutes.

4.6.2.3 Wildcat hand brake test. The wildcat brake shall be subjected to a wildcat brake torque capacities test to demonstrate that the equipment can resist the calculated breaking torque without deformation or damage. The torque applied shall be relative to a 15-fathom drop of the anchor and chain. The wildcat brake shall be set using the handwheel with the drive motor brake released. A calibrated tension device shall be used to ensure the load required to the anchor rigging and tension device. After the load is stabilized, the time duration shall be a minimum of 10 minutes.

4.6.2.4 Electric motor brake test. The test of motor brake torque capacities shall demonstrate that the equipment can resist the calculated torque without deformation or damage. The drive unit brake shall stop and hold 150 percent of the combined weight of the anchor and 80 fathoms of chain while being lowered at full speed under control of the drive unit. A calibrated tension device for the load required shall induce a tension load to the anchor rigging and tension device. After the load is stabilized, the time duration shall be a minimum of 10 minutes. As an alternative to this physical test, dynamic simulation of the electric brake may be used to simulate one cycle of powered operation, raising and lowering the anchor, and paying out of 100 fathoms of chain in a water column of 100 fathoms.

4.6.2.5 Wildcat chain testing. The five-link and three-link sections shall be removed from the bitter end of the anchor chain. The five-link and three-link sections shall be reinstalled on the bitter end of the anchor chain after completing a wildcat fit check using the detachable link furnished for the test.

4.6.3 Dock tests. Each windlass shall be tested dockside after installation on board the ship as specified herein. Ships with two windlasses shall perform these tests with both windlasses for both anchors (“crossover” operation).

4.6.3.1 No-load test (disengaged and engaged). Each windlass unit shall be run for 30 minutes in both directions with the wildcat disengaged. The no-load engaged test shall be with the wildcat engaged, and there shall be no added load on the anchor. It shall be confirmed that the anchor releases from the housed position by using the band brake. Additionally, it shall be demonstrated that the band brake can stop the anchor. The anchor shall run free approximately to the waterline prior to fully stopping.

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4.6.3.2 Normal operation test. Each windlass unit shall be tested by hoisting and lowering the anchor to 30 fathoms in 33 fathoms or more of water at an average rate of not less than 6 fathoms per minute continuously for four complete cycles. This test may be conducted in a lesser depth of water, provided that sufficient weights are added so that the total weight lifted is equivalent to the weight of the anchor and 30 fathoms of chain in the water. This test shall also cover inspection of the operation of the chain on the wildcat and in the chain pipe. This test shall be repeated (four cycles) with a total weight equivalent to the weight of the anchor and 60 fathoms of chain. Prior to each evolution, confirmation shall be made that the chain swivel is functioning as intended and is free to rotate.

4.6.3.2.1 100-fathom test. Without exceeding the full-load rating of the motor, each windlass shall be tested by hoisting a weight equal to 100 fathoms of chain plus the anchor weight (with compensatory weight added [as described above]) a distance of 30 fathoms at no specified speed.

4.6.3.2.2 Emergency run test (thermal overload) simulation. Local and remote operation of controls and emergency run capability shall be demonstrated. Emergency run capability shall be demonstrated at low and high speed, with low speed defined as 1 fathom per minute, and high speed defined as 6 fathoms per minute. This test shall also verify 3.9 through 3.9.3.

4.6.3.3 Wildcat hand brake test. Each unit shall be tested with weight added as described above so that the total weight is equivalent to the weight of the anchor and 30 fathoms of chain in the water. The anchor shall be held in the stowage position in the hawsepipe by the hand brake while the locking head is disconnected. The brake shall then be released and the anchor shall be let go until 15 fathoms of chain are out or until the anchor is within 2 fathoms of the bottom in testing conditions where the water is less than 15 fathoms deep, at which time the brake shall be applied. The brake shall stop and hold the anchor and chain within 2 fathoms travel or before the anchor grounds. The brake test shall be repeated until the wildcat hand brake demonstrates successful operation.

4.6.3.4 Electric motor brake test. For electric motor brake testing, a weight equal to 100 fathoms of chain plus the anchor weight shall be lowered as in normal operation until the anchor is within 3 fathoms of the bottom, at which time the electric motor shall be de-energized, and the electric brake shall function. The electric brake shall stop and hold the load before the anchor grounds.

4.6.3.5 Stowage (bolstering) test. The anchor shall stow at no greater than a speed of 1 fathom per minute. The anchor shall make a minimum of three-point contact with the hull of the ship. Three-point contact is defined as three points of contact between the hull of the ship and the crown feet or flukes of the anchor. The anchor shall not slide or pound against the side of the ship.

4.6.3.6 Capstan load tests. Each capstan head shall be tested statically and at normal load in accordance with 4.6.3.6.1 and 4.6.3.6.2.

4.6.3.6.1 Static test. Each capstan head shall be tested statically by using the head as a bitt to hold a load equal to the breaking strength of the specified synthetic rope (see 6.2) for 10 minutes with no deformation allowed. The capstan driving mechanism is not required to raise a weight equal to the static rating.

4.6.3.6.2 Normal load test. Each capstan head with the wildcat disconnected shall be used to demonstrate by tension dynamometer or by other means that the capstan is capable of pulling its rated load at the rated speed. No part of the equipment shall show any sign of permanent distortion or undue wear after these tests.

4.6.4 Deep sea tests. When the ship is on sea trials and in a depth of between 65 fathoms but not exceeding 70 fathoms, anchor-handling arrangements and gear shall be tested as specified herein to demonstrate satisfactory operation in all respects under service conditions.

4.6.4.1 Hoisting test. With the anchor bending shackle just outside the hawsepipe, each anchor shall be independently lowered under power until 60 fathoms of chain are out of the hawsepipe and then hoisted to the stowage position. Lowering and hoisting of the anchors and chain to 60 fathoms shall be at an average rate of not less than 6 fathoms per minute. The anchor shall not ground during evolution.

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4.6.4.2 Wildcat hand brake test. With the anchor in the hawsepipe and with the locking head disconnected, the anchor shall be dropped by releasing the hand brake. Anchor descent shall be stopped at 15, 30, 45, and 60 fathoms of chain out of the hawsepipe by operating the brake handwheel without additional leverage. Testing shall show that the anchor releases from the hawsepipe in the stowed position when the band brake is released. The anchor shall not need assistance to start motion prior to free fall operation. In performing this test, the brake shall be freed by turning the handwheel slowly to the position where the anchor begins its free fall. From this position, after a delay of approximately 2 seconds, the brake shall be rapidly reset by the time the specified fathom mark reaches the hawsepipe. The brake shall stop and hold the anchor and chain in not more than 2 fathoms of travel with 15 and 30 fathoms out, in not more than 3 fathoms of travel with 45 fathoms out, and in not more than 4 fathoms of travel with 60 fathoms out. During each test (i.e., 0 to 15 fathoms, 15 to 30 fathoms, 30 to 45 fathoms, or 45 to 60 fathoms), the system shall not come to a complete stop at any time during the test. Each test shall be completed in one full evolution. During testing, unexpected chain motion (i.e., chain rotation, chain skipping in the wildcat, or chain not loading correctly into the wildcat) is unacceptable. The anchor shall not ground during evolution.

4.6.4.3 Electric brake test. Electric brakes shall be tested while lowering the anchor at the maximum speed under power. When 60 fathoms of chain have been let out, the electric motor shall be de-energized, and the electric brake shall function. The brake shall stop and hold the anchor. The length of chain payed out after the brake is energized shall be recorded. The anchor shall not ground during evolution.

4.6.4.4 Chain handling test. The installed windlass shall be used to rig the ship to be towed in accordance with S9086-TW-STM-010/582. The windlass shall demonstrate the capability to handle the chain while executing this rigging.

4.6.5 Reliability and maintainability test.

4.6.5.1 Reliability. The anchor windlass unit shall meet the requirements of 3.6. If required (see 6.2), new designs or major redesigns shall, at a minimum, have a failure mode, effects, and criticality analysis performed. During testing, a failure reporting, analysis, and corrective action, a system shall be used to record failures and document corrective actions. The duration of a test and the allowable number of failures for the purpose of demonstrating a mean time between failure of 1,000 cycles and statistical confidence of 80 percent (or as specified in 6.2) shall be in accordance with table III.

TABLE III. Reliability test data.

Duration of demonstration test (hours)	Maximum allowable failures
1,600	0
3,000	1
4,300	2
5,500	3

One complete cycle comprises lowering an anchor and chain 60 fathoms at a speed of 1,000 feet per minute followed by hoisting the anchor and chain at a rate of 6 fathoms per minute or approximately 15 minutes per cycle. The anchor and chain can be replaced with a simulated load placed on the capstan head. Parts and components that are subject to wearing out can be replaced at cyclic intervals consistent with the manufacturer's recommendations. A failure in this context is any event that results in the need for corrective maintenance or repair in order for the anchor windlass unit to lower and raise the anchor and chain or simulated load.

4.6.5.2 Maintainability. The maintenance procedures, removal of parts for repair, and the time required for accomplishment shall be verified during first article testing. The demonstration procedures in MIL-HDBK-470 may be used.

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4.6.6 Software test. Software shall be tested as specified (see 6.2).

4.6.7 Cybersecurity. Cybersecurity requirements shall be verified in accordance with MIL-DTL-32613. PLCs shall, additionally, have cybersecurity requirements verified in accordance with MIL-PRF-32006.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The electro-mechanical windlass is intended for use in lowering and raising an anchor and chain when anchoring and to reduce the use of hydraulic machinery aboard naval ships. The capstan head is intended for use in handling lines and for warping service. Further use of anchor windlasses procured to prior revisions of this specification should be discontinued and all future requisitions should be to the current revision of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of anchor windlass, including capstan option (see 1.2).

c. When first article inspection is required (see 3.1).

d. When cast iron is to be used, if required (see 3.3).

e. Environmental condition requirements, if other than specified herein (see 3.4 and 4.4).

f. Acquisition information and data requirements as identified in MIL-DTL-901, as applicable, and any additional shock requirements, if necessary (see 3.4.2).

g. Weight of anchor to be handled (see 3.5, 3.5.a, and 3.7.1).

h. Maximum permissible size and weight of windlass, if required (see 3.5.a and 3.10).

i. Allowable stress loads, if other than specified herein (see 3.5.a).

j. Synthetic rope size and strength to be used by capstan, if capstan is specified (see 3.5.a, 3.5.2.2, 3.5.4, 3.7.2, 3.10, and 4.6.3.6.1).

k. Chain size to be handled (see 3.5.a, 3.5.3, 3.7.1, and 3.10).

l. Design considerations for dual windlass installations (see 3.5.d).

m. Speed limiting device, when required (see 3.5.3.6).

n. Remote operability requirements, if necessary (see 3.5.6.d).

o. Bedplate fairleads, if required (see 3.5.7.1).

p. Chain pipe and deck bolsters details (see 3.5.7.1).

q. Power-assist and electric motor brakes, if necessary (see 4.6.3.4).

r. Software and software testing requirements (see 3.5.10, 4.6.6, and 6.4).

s. Reliability and maintainability demonstration and analysis requirements as well as additional ship specific requirements (if necessary) (see 3.6 and 4.6.5.1).

t. Additional coating requirements, if other than specified herein (see 3.11.1).

u. Packaging requirements (see 5.1).

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v. Additional documentation (see 6.3).

w. Software deliverables (see 6.4).

6.3 Recommended documentation. When specified (see 6.2), the following documentation should be requested as part of the acquisition of an anchor windlass unit:

a. Test reports.

b. Calculations.

(1) All calculations, assumptions, and variables used when designing all system components in all planned load conditions in both inhaul and launch evolutions should be provided. This includes, but is not limited to, defining heat requirements, material and sizing requirements, electric brake requirements, motor requirements combine stress requirements, gearbox design sizing (including AGMA variables used in design), and the values should be provided in the calculation package. AGMA calculations should be compared to tensile yield requirements. Calculations or testing should prove that, during normal bolstering operations at 1 fathom per minute, stresses do not exceed 70 percent, and that bolstering at any speed should not exceed 100 percent of the tensile yield point of the ductile material. Calculations should show that a duty cycle time out does not occur in any planned anchoring condition.

(2) When powered assist is not utilized, calculations showing that powered assist is not needed for operation by a 5 percent female in accordance with MIL-STD-1472 over the life of ship.

(3) Calculations should be performed to show that in a two-man tight condition, the brake should not slip prior to the time needed to enable the installation of the chain stoppers.

(4) If analysis is used to complete reliability or maintainability analysis, assumptions used and results of analysis.

c. Technical manuals.

d. Product drawings.

e. Provisioning technical documentation.

6.4 Software. When specified (see 6.2), contracting officers should identify any and all software requirements for operation, maintenance, and modification of the anchor windlass system, to include the following.

6.4.1 Programming. The programming, software, firmware, and user programs should be provided in electronic format and should be compatible with use on a portable computer for the purpose of viewing and editing programs and for uploading and downloading programs to and from the system via a serial ethernet communication port. There should be no proprietary protocols or communication ports. The source code for all software should be provided. All necessary programming tools and communication cables should be provided.

6.4.2 Program access control. Software and programs should be capable of incorporating password-based access controls to limit the upload, download, deletion, or modification of software and programs to personnel inputting the correct authorization password. An ethernet communication port should be provided on the controller enclosure via a lockable, hinged cover to allow for programming, monitoring, and troubleshooting of the programmable control system with the enclosure door closed.

6.4.3 Program retention. All software and user programs necessary to operate, monitor, and maintain the system should be stored in non-volatile memory within the system. Removal or interruption of electrical power should not cause software and user programs to be erased, altered, or corrupted. Energy storage devices should not be used to meet this requirement.

6.5 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

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6.6 Subject term (key word) listing.

Band brake

Capstan

Chain

Cybersecurity

Deck machinery

Electric brake

Variable-speed drive (VSD)

Wildcat

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activities:
Navy – CG
DLA – CC

Preparing activity:
Navy – SH
(Project 3950-2022-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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